Area Under Curve

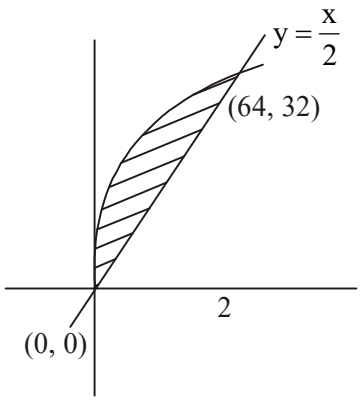
149. The area of the region $\left\{(x,y):|x-y|\le y\le 4\sqrt{x}\right\}$:

क्षेत्र $\left\{(x,y):|x-y|\le y\le 4\sqrt{x}\right\}$ का क्षेत्रफल है :

(A) $\frac{1024}{3}$ (B) 512 (C) $\frac{2048}{3}$ (D) $\frac{512}{3}$

Ans. A

Sol.



$|x-y|\le y\le 4\sqrt{x}$

Now, $y=|x-y|$

$y^2=(x-y)^2$

$\Rightarrow y=\frac{x}{2}$ and $x=0$

Now, area $=\int_0^{64}\left(4\sqrt{x}-\frac{x}{2}\right)dx$

$$=\left[\frac{4x^{\frac{3}{2}}}{\frac{3}{2}}-\frac{x^2}{4}\right]_0^{64}=\frac{8}{3}\cdot 8^3-\frac{64^2}{4}=64^2\left(\frac{1}{12}\right)$$

$$=\frac{1024}{3}$$

150. A line passing through the point A(–2, 0), touches the parabola P: $y^2=x-2$ at the point B in the first quadrant. The area, of the region bounded by the line AB, parabola P and the x-axis, is :

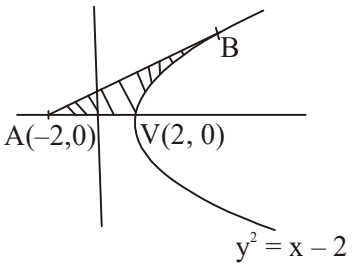
बिन्दु A(–2, 0) से होकर जाने वाली एक रेखा परवलय P : $y^2 = x - 2$ को प्रथम चतुर्थांश में बिन्दु B पर स्पर्श करती है। रेखा AB परवलय P और x-अक्ष से परिबद्ध क्षेत्र का क्षेत्रफल है :

(A) 2 (B) 3 (C) $\frac{8}{3}$ (D) $\frac{7}{3}$

Ans. C

Sol. Equation of tangent

$$y = m(x-2) + \frac{\frac{1}{4}}{m}$$



Passes through (–2, 0)

$$0 = -4m + \frac{1}{4m}$$

$$\Rightarrow m = \frac{1}{4}$$

$$y = \frac{1}{4}(x-2) + 1$$

$x - 4y + 2 = 0$

Solving with $y^2 = x - 2$,

B(6, 2)

$$\text{Area} = \int_0^2 \left((y^2+2) - (4y-2)\right) dy = \frac{8}{3}$$

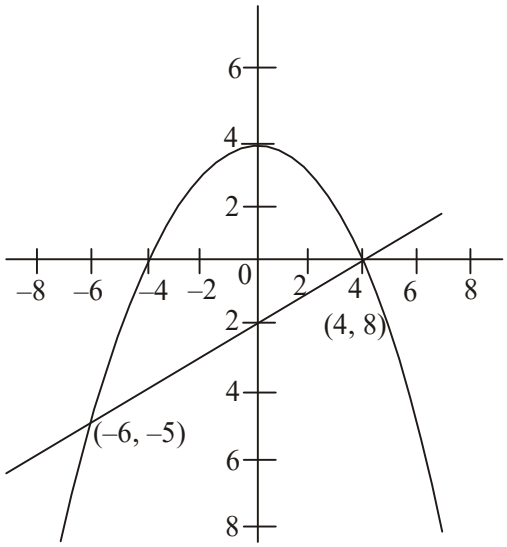
151. If the area of the region bounded by the curves $y = 4 - \frac{x^2}{4}$ and $y = \frac{x-4}{2}$ is equal to α, then 6α equals :

यदि वक्रों $y = 4 - \frac{x^2}{4}$ और $y = \frac{x-4}{2}$ से परिबद्ध क्षेत्र का क्षेत्रफल α है, तो 6α बराबर है :

(A) 250 (B) 220 (C) 240 (D) 210

Ans. A

Sol.



$$\text{Area} = \int_{-6}^{4} \left\{ \left(4 - \frac{x^2}{4}\right) - \left(\frac{x-4}{2}\right) \right\} dx$$

$$= \int_{-6}^{4} \left\{ -\frac{x^2}{4} - \frac{x-6}{2} \right\} dx$$

$$\alpha = -\frac{x^3}{12} - \frac{x^2}{4} + 6x \Big|_{-6}^{4} = \frac{125}{3}$$

$$\therefore 6\alpha = 250$$

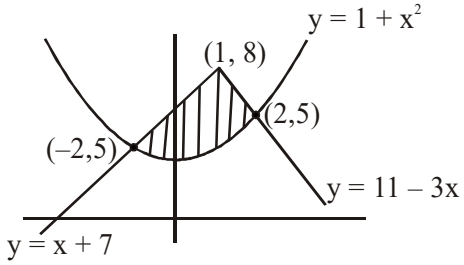
152. If the area of the region $\{(x, y) : 1 + x^2 \le y \le \min \{x + 7, 11 - 3x\}\}$ is A, then 3A is equal to :

यदि क्षेत्र $\{(x, y) : 1 + x^2 \le y \le \min \{x + 7, 11 - 3x\}\}$ का क्षेत्रफल A है, तो 3A बराबर है :

(A) 46 (B) 47 (C) 49 (D) 50

Ans. D

Sol.



$$A = \int_{-2}^{1}\left(\left(x+7\right)-\left(x^2+1\right)\right)dx + \int_{1}^{2}\left(\left(11-3x\right)-\left(x^2+1\right)\right)dx$$

$$= \left(-\frac{x^3}{3}+\frac{x^2}{2}+6x\right)_{-2}^{1} + \left(-\frac{x^3}{3}-\frac{3x^2}{2}-10x\right)_{1}^{2}$$

$$= \frac{50}{3}$$

So, 3A = 50

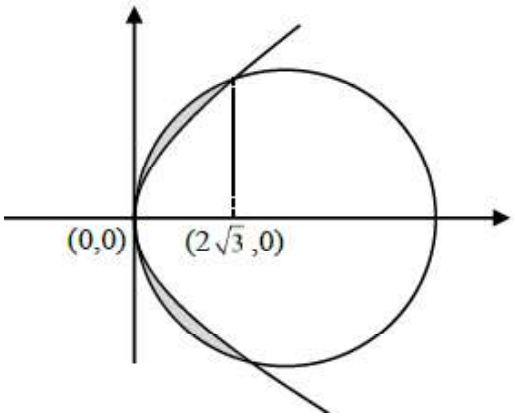
153. The area of the region, inside the circle $\left(x-2\sqrt{3}\right)^2+y^2=12$ and outside the parabola $y^2=2\sqrt{3}x$ is :

वृत्त $\left(x-2\sqrt{3}\right)^2+y^2=12$ के अन्दर और परवलय $y^2=2\sqrt{3}x$ के बाहर के क्षेत्र का क्षेत्रफल है :

(A) $3\pi+8$ (B) $6\pi-16$ (C) $3\pi-8$ (D) $6\pi-8$

Ans. B

Sol.



$$y^2 = 2\sqrt{3}x$$

$$\left(x-2\sqrt{3}\right)^2+y^2=\left(2\sqrt{3}\right)^2$$

$$A = \frac{\pi r^2}{2} - 2\int_0^{2\sqrt{3}} \sqrt{2\sqrt{3}x}\,dx$$

$$\frac{\pi(12)}{2} - 2\sqrt{2\sqrt{3}}\,\frac{\left(x^{3/2}\right)_0^{2/3}}{3/2}$$

$= 6\pi - 16$

154. Let $f: R \to R$ be a twice differentiable function such that $f(x + y) = f(x)\,f(y)$ for all $x, y \in R$. If $f'(0) = 4a$ and f satisfies $f''(x) - 3af'(x) - f(x) = 0$, $a > 0$, then the area of the region $R = \{(x, y) \mid 0 \le y \le f(ax), 0 \le x \le 2\}$ is:

माना कि $f: R \to R$, एक दो बार अवकलनीय फलन इस प्रकार है कि सभी $x, y \in R$ के लिए, $f(x + y) = f(x)\,f(y)$ है। यदि $f'(0) = 4a$ है और f, $f''(x) - 3af'(x) - f(x) = 0$, $a > 0$, को संतुष्ट करता है, तो क्षेत्र $R = \{(x, y) \mid 0 \le y \le f(ax), 0 \le x \le 2\}$ का क्षेत्रफल है :

(A) $e^2 + 1$ (B) $e^4 + 1$ (C) $e^2 - 1$ (D) $e^4 - 1$

Ans. C

Sol. $f(x+y) = f(x)\cdot f(y)$

$\Rightarrow f(x) = e^{\lambda x}$ $f'(0) = 4a$

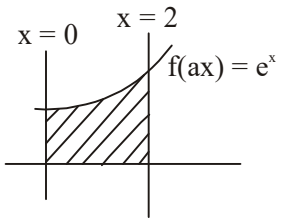
$\Rightarrow f'(x) = \lambda e^{\lambda x} \Rightarrow \lambda = 4a$

So, $f(x) = e^{4ax}$

$f''(x) - 3af'(x) - f(x) = 0$

$\Rightarrow \lambda^2 - 3a\lambda - 1 = 0$

$\Rightarrow 16a^2 - 12a^2 - 1 = 0 \Rightarrow 4a^2 = 1 \Rightarrow a = \frac{1}{2}$



$F(x) = e^{2x}$

$$\text{Area} = \int_0^2 e^x dx = e^2 - 1$$

155. The area of the region enclosed by the curves $y = x^2 - 4x + 4$ and $y^2 = 16 - 8x$ is :

वक्रों $y = x^2 - 4x + 4$ और $y^2 = 16 - 8x$ द्वारा परिबद्ध क्षेत्र का क्षेत्रफल है :

(A) $\frac{4}{3}$ (B) 8 (C) 5 (D) $\frac{8}{3}$

Ans. D

Sol.

$y = x^2 - 4x + 4$	$y^2 = 16 - 8x$
$(x-2)^2 = y$	$y^2 = -8(x-2)$
$X^2 = Y$	$Y^2 = -8X$

$$\text{Now Area} = \frac{(4a)(4b)}{3} = \frac{8\times 1}{3} = \frac{8}{3}$$

156. If the area of the region $\{(x, y) : -1 \le x \le 1, 0 \le y \le a + e^{|x|} - e^{-x}, a > 0\}$ is $\frac{e^2+8e+1}{e}$, then the value of a is:

यदि क्षेत्र $\{(x, y) : -1 \le x \le 1, 0 \le y \le a + e^{|x|} - e^{-x}, a > 0\}$ का क्षेत्रफल $\frac{e^2+8e+1}{e}$ है, तो a का मान है :

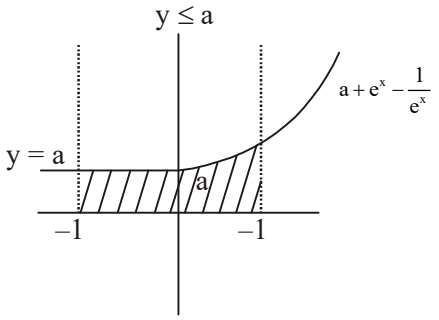
(A) 7 (B) 8 (C) 5 (D) 6

Ans. C

Sol. $\{(x, y) : -1 \le x \le 1, 0 \le y \le a + e^{|x|} - e^{-x}, a > 0\}$

Now, $y \le a + e^{|x|} - e^{-x}$

C-I : $x \ge 0$	C-II : $x < 0$
$y \le a + e^x - e^{-x}$	$y \le a + e^{-x} - e^{-x}$
$y \le a + e^x - \frac{1}{e^x}$	$y \le a$



$\therefore$ A.T.Q

$$\int_{-1}^{0} a\,dx + \int_{0}^{1} \left(a + e^x - e^{-x}\right) dx = \frac{e^2+8e+1}{e}$$

$ax\Big|_{-1}^{0} + ax + e^{x} + e^{-x}\Big|_{0}^{1}$

$a[0-(-1]+(a+e+e^{-1})-(2)=\dfrac{e^2+8e+1}{e}$

$2a+e+\dfrac{1}{e}-2=\dfrac{e^2+8e+1}{e}$

$2a=\dfrac{e^2+8e+1}{e}-\left(\dfrac{e^2-2e+1}{e}\right)$

$=\dfrac{e^2+8e+1-e^2+2e-1}{e}$

$a = 5$

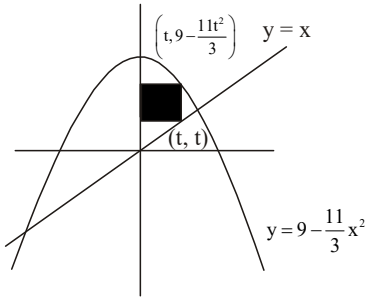
157. Consider the region $R=\left\{(x,y): x \le y \le 9-\dfrac{11}{3}x^2, x \ge 0\right\}$. The area, of the largest rectangle of sides parallel to the coordinate axes and inscribed in R, is :

क्षेत्र $R=\left\{(x,y): x \le y \le 9-\dfrac{11}{3}x^2, x \ge 0\right\}$ पर विचार करें। निर्देशांक अक्षों के समान्तर भुजाओं वाले और R के अंतर्गत अंकित सबसे बड़े आयत का क्षेत्रफल है :

(A) $\dfrac{821}{123}$ (B) $\dfrac{730}{119}$ (C) $\dfrac{625}{111}$ (D) $\dfrac{567}{121}$

Ans. D

Sol.



$A = t\cdot\left(9-\dfrac{11t^2}{3}-t\right)$

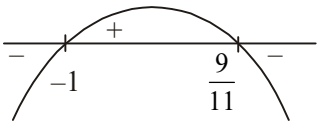
$A=9t-t^2-\dfrac{11}{3}t^3$

$\frac{dA}{dt} = 9 - 2t - 11t^2$

$\Rightarrow 11t^2 + 2t - 9 = 0$

$11t^2 + 11t - 9t - 9 = 0$

$t = -1 \ \& \ t = \frac{9}{11}$

$\therefore \frac{dA}{dt} =$ 

$\downarrow$

Maxima at $t = \frac{9}{11}$

$\therefore$ largest area $= \frac{9}{11}\left(9 - \frac{11}{3}, \frac{81}{121} - \frac{9}{11}\right)$

$= \frac{9}{11} \cdot \frac{63}{11} = \frac{567}{121}$

158. The area of region $\{(x, y) : x^2 + 4x + 2 \le y \le |x + 2|\}$ is equal to :

क्षेत्र $\{(x, y) : x^2 + 4x + 2 \le y \le |x + 2|\}$ का क्षेत्रफल बराबर है :

(A) 5 (B) 20/3 (C) 7 (D) 24/5

Ans. B

Sol. $x^2 + 4x + 2 \le y \le |x + 2|$

The area bounded between

$y = x^2 + 4x + 2 = (x + 2)^2 - 2$

and $y = |x + 2|$ is same as area bounded between $y = x^2 - 2$ and $y = |x|$

For Point of Intersection $|x|^2 - 2 = |x|$

$\Rightarrow |x| = 2 \Rightarrow x = \pm 2$

$\therefore$ Required area $= -\int_{-2}^{2} \left(x^2 - 2\right) dx + \int_{-2}^{2} |x| dx$

$= -2\int_{0}^{2} \left(x^2 - 2\right) dx + 2\int_{0}^{2} x.dx$

$$= -2\left[\frac{x^3}{3} - 2x\right]_0^2 + 2\left[\frac{x^2}{2}\right]_0^2$$

$$= -2\left[\frac{8}{3} - 4\right] + 2\left[\frac{4}{2}\right]$$

$$= -2 \times \left(\frac{-4}{3}\right) + 4$$

$$= \frac{20}{3}$$

159. The area of the region enclosed by the curves $y = e^x$, $y = |e^x - 1|$ and y-axis is :

वक्रों $y = e^x$, $y = |e^x - 1|$ और y-अक्ष से घिरे क्षेत्र का क्षेत्रफल है :

(A) $\log_e 2$ (B) $1 + \log_e 2$ (C) $1 - \log_e 2$ (D) $2\log_e 2 - 1$

Ans. C

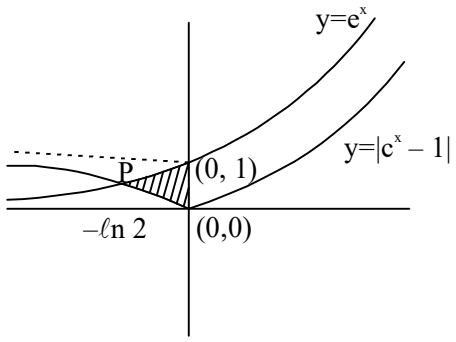
Sol. P : $e^x = -(e^x - 1)$

$e^x = 1/2$

$x = -\ell n 2$

$$\text{Area} = \int_{-\ell n 2}^{0} e^x - \left(-e^x + 1\right)$$

$$= \left(2e^x - x\right)_{-\ell n 2}^{0} = 1 - \ell n 2$$



160. The area (in sq. units) of the region $\{(x, y) : 0 \le y \le 2|x| + 1, 0 \le y \le x^2 + 1, |x| \le 3\}$ is :

क्षेत्र $\{(x, y) : 0 \le y \le 2|x| + 1, 0 \le y \le x^2 + 1, |x| \le 3\}$ का क्षेत्रफल (वर्ग इकाई में) है :

(A) $\frac{64}{3}$ (B) $\frac{17}{3}$ (C) $\frac{32}{3}$ (D) $\frac{80}{3}$

Ans. A

Sol. Required area $= 2\left[\int_0^2 \left(x^2+1\right)dx + \int_2^3 (2x+1)\,dx\right]$

$= \frac{64}{3}$

161. The area of the region bounded by the curves $x(1 + y^2) = 1$ and $y^2 = 2x$ is :

वक्रों $x(1 + y^2) = 1$ और $y^2 = 2x$ द्वारा परिबद्ध क्षेत्र का क्षेत्रफल है :

(A) $2\left(\frac{\pi}{2}-\frac{1}{3}\right)$ (B) $\frac{1}{2}\left(\frac{\pi}{2}-\frac{1}{3}\right)$ (C) $\frac{\pi}{4}-\frac{1}{3}$ (D) $\frac{\pi}{2}-\frac{1}{3}$

Ans. D

Sol. $C_1 : \left(1+y^2\right) = \left(\frac{1}{x}\right)$

$y^2 = \left(\frac{1}{x}-1\right)$

$C_2 : y^2 = 2x$

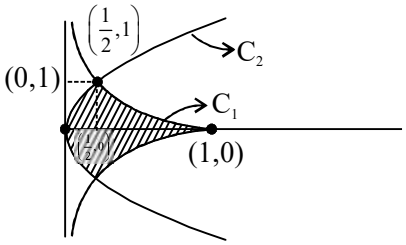
$\left(\frac{1-x}{x}\right) = 2x$

$2x^2 = 1 - x$

$2x^2 + x - 1 = 0$

$(2x - 1)(x + 1) = 0$

$x = \frac{1}{2}, y = 1$



Resultant area with respect to y-axis $= 2\int_0^1 \left[\left(\frac{1}{1+y^2}\right)-\left(\frac{y^2}{2}\right)\right]\cdot dy$

$\Rightarrow 2\left[\tan^{-1}(y) - \frac{y^3}{6}\right]_0^1$

$\Rightarrow 2\left[\frac{\pi}{4}-\frac{1}{6}\right]$

$\Rightarrow\left(\frac{\pi}{2}-\frac{1}{3}\right)$

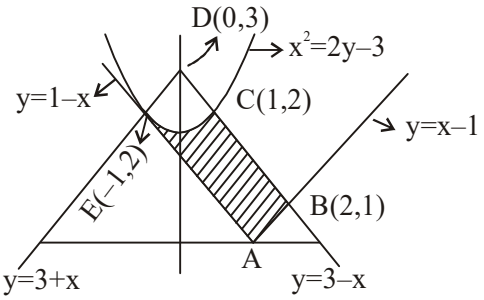
162. Let the area of the region $\left\{(x,y): 2y \le x^2+3, y+|x| \le 3, y \ge |x-1|\right\}$ be A. Then 6A is equal to :

माना कि क्षेत्र $\left\{(x,y): 2y \le x^2+3, y+|x| \le 3, y \ge |x-1|\right\}$ का क्षेत्रफल A है। तब 6A बराबर है :

(A) 16 (B) 18 (C) 12 (D) 14

Ans. D

Sol.



A = area of rectangle ABDE – area of region EDC

$$A = 4 - 2\int_0^1 (3-x) - \left(\frac{x^2+3}{2}\right)dx$$

$$A = 4 - 2\left(3 - \frac{1}{2} - \frac{1}{6} - \frac{3}{2}\right) = \frac{7}{3}$$

6A = 14

163. Let the area of enclosed between the curves $|y| = 1 - x^2$ and $x^2 + y^2 = 1$ be α. If $9\alpha = \beta\pi + \gamma$; β, γ are integers, then the value of $|\beta - \gamma|$ equals :

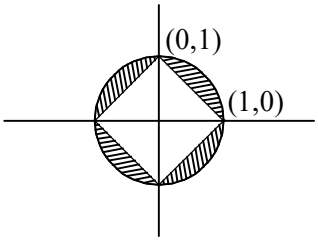
माना कि वक्रों $|y| = 1 - x^2$ और $x^2 + y^2 = 1$ के बीच घिरे क्षेत्र का क्षेत्रफल α है। यदि $9\alpha = \beta\pi + \gamma$; β, γ पूर्णांक है, तो $|\beta - \gamma|$ का मान बराबर है :

(A) 18 (B) 33 (C) 27 (D) 15

Ans. B

Sol. $C_1 : |y| = 1 - x^2$

$C_2 : x^2 + y^2 = 1$



$\therefore$ Required Area

$= \alpha = 4[\text{Area of circle in } 1^{st} \text{ quad. } -\int_0^1 \left(1-x^2\right)dx\ ]$

$= 4\left[\frac{\pi}{4} - \left[x - \frac{x^3}{3}\right]_0^1\right]$

$\alpha = \pi - \frac{8}{3}$

$\therefore\ 3\alpha = 3\pi - 8$

$\therefore\ 9\alpha = 9\pi - 24$

$\therefore\ \beta = 9, \gamma = -24$

$\therefore\ |\beta - \gamma| = 33$

164. One of the points of intersection of the curves $y = 1 + 3x - 2x^2$ and $y = \frac{1}{x}$ is $\left(\frac{1}{2}, 2\right)$. Let the area of the region enclosed by the curves be $\frac{1}{24}\left(\ell\sqrt{5} + m\right) - n\log_e\left(1+\sqrt{5}\right)$, where $\ell, m, n \in N$. Then $\ell + m + n$ is equal to :

वक्रों $y = 1 + 3x - 2x^2$ तथा $y = \frac{1}{x}$ का एक प्रतिच्छेदन $\left(\frac{1}{2}, 2\right)$ है। माना इन वक्रों से घिरे का क्षेत्रफल $\frac{1}{24}\left(\ell\sqrt{5} + m\right) - n\log_e\left(1+\sqrt{5}\right)$ है, जहाँ $\ell, m, n \in N$ है, तो $\ell + m + n$ का मान होगा :

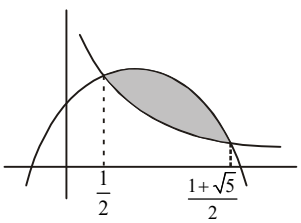
(A) 32 (B) 29 (C) 31 (D) 30

Ans. D

Sol. $2x^3 - 3x^2 - x + 1 = 0$

$\Rightarrow (2x - 1)(x^2 - x - 1) = 0$

$\Rightarrow\ x = \frac{1}{2}, x = \frac{1+\sqrt{5}}{2}$



$$\text{Area} = \int_{\frac{1}{2}}^{\frac{1+\sqrt{5}}{2}} \left(1+3x-2x^2-\frac{1}{x}\right)dx$$

$$= \left[x + + \frac{3x^2}{2} - \frac{2x^3}{3} - \ell n x\right]$$

$$= \frac{\sqrt{5}+1}{2} + \frac{3}{8}\left(\sqrt{5}+1\right)^2 - \frac{1}{12}\left(\sqrt{5}+1\right)^3 - \ell n\left(\frac{\sqrt{5}+1}{2}\right) - \left(\frac{1}{2}+\frac{3}{8}-\frac{1}{12}-\ell n\frac{1}{2}\right)$$

$$= \frac{1}{24}\left[12\left(\sqrt{5}+1\right)+9\left(\sqrt{5}+1\right)^2-2\left(\sqrt{5}+1\right)^3-12-9+2\right] - \ell n\left(\frac{\sqrt{5}+1}{2}\times 2\right)$$

$$= \frac{1}{24}\left[12\left(\sqrt{5}+1\right)+9\left(6+2\sqrt{5}\right)-2\left(5\sqrt{5}+1+3\sqrt{5}\left(\sqrt{5}+1\right)-19\right)\right] - \ell n\left(\sqrt{5}+1\right)$$

$$= \frac{1}{24}\left[14\sqrt{5}+15\right] - \ell n\left(\sqrt{5}+1\right)$$

$\therefore l = 14, m = 15, n = 1$

$l + m + n = 30$

165. The area enclosed between the curves $y = x|x|$ and $y = x - |x|$ is :

वक्रों $y = x|x|$ तथा $y = x - |x|$ के बीच घिरे क्षेत्र का क्षेत्रफल है :

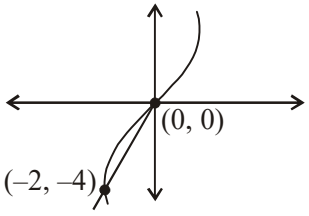
(A) 2/3 (B) 4/3 (C) 8/3 (D) 1

Ans. B

Sol. $y = x|x|$ & $y = x - |x|$

$$y = x|x| = \begin{cases} x^2 & , x > 0 \\ -x^2 & , x < 0 \end{cases}$$

$$y = x - |x| = \begin{cases} 0 & , x > 0 \\ 2x & , x < 0 \end{cases}$$



point of intersections are (0 , 0) & (–2, 4).

$$\therefore \text{Area} = \int_{-2}^{0} \left(-x^2-2x\right)dx$$

$$= \left[\frac{-x^3}{3} - \frac{2x^2}{2}\right]_{-2}^{0}$$

$$= -\left[\frac{8}{3} - 4\right] = \frac{4}{3} \text{ sq. unit}$$

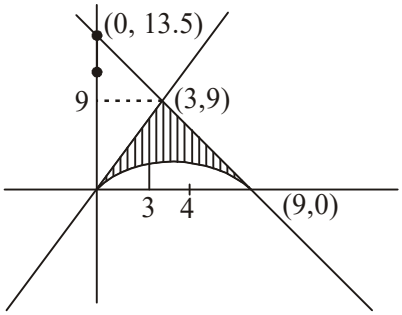
166. Let the area of the region enclosed by the curves $y = 3x$, $2y = 27 - 3x$ and $y = 3x - x\sqrt{x}$ be A. Then 10A is equal to :

माना वक्रों $y = 3x$, $2y = 27 - 3x$ और $y = 3x - x\sqrt{x}$ से घिरे क्षेत्र का क्षेत्रफल A है, तो 10A बराबर है :

(A) 184 (B) 172 (C) 154 (D) 162

Ans. D

Sol.



$$y = 3x - x^{\frac{3}{2}} \qquad y' = 3 - \frac{3}{2}\sqrt{x}$$

$$y' = 3\left(1 - \frac{\sqrt{x}}{2}\right)$$

$f'(x)$ + −

Area of shaded region

$$= \int_0^3 \left(3x - \left(3x - x^{\frac{3}{2}}\right)\right)dx + \int_3^9 \frac{27 - 3x}{2} - \left(3x - x^{\frac{3}{2}}\right)dx$$

$$= \int_0^3 x^{\frac{3}{2}}dx + \frac{1}{2}\int_3^9 \left(27 - 9x + 2x^{\frac{3}{2}}\right)dx = \left(\frac{2}{5}x^{\frac{5}{2}}\right)_0^3 + \frac{1}{2}\left[27x - \frac{9x^2}{2} + 2\times\frac{2}{5}x^{\frac{5}{2}}\right)_3^9$$

$$= \frac{2}{5}\times 9\sqrt{3} + \frac{1}{2}\left[243 - \frac{729}{2} + \frac{4}{5}\times 81\times 3 - 81 + \frac{81}{2} - \frac{4}{5}\times 9\sqrt{3}\right]$$

$$= \frac{1}{2}\left[\frac{486-729-81}{2}+\frac{972}{5}\right] = \frac{81}{5}$$

$\Rightarrow 10A = 162$

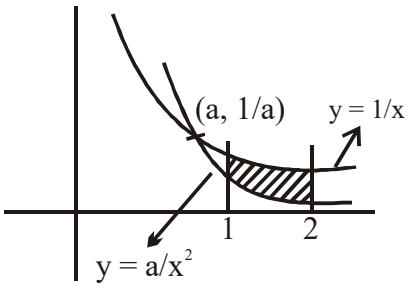
167. If the area of the region $\left\{(x,y): \frac{a}{x^2} \le y \le \frac{1}{x}, 1 \le x \le 2, 0 < a < 1\right\}$ is $\left(\log_e 2\right) - \frac{1}{7}$ then the value of $7a-3$ is equal to :

यदि क्षेत्र $\left\{(x,y): \frac{a}{x^2} \le y \le \frac{1}{x}, 1 \le x \le 2, 0 < a < 1\right\}$ का क्षेत्रफल $\left(\log_e 2\right) - \frac{1}{7}$ है, तो $7a - 3$ का मान बराबर है :

(A) 1 (B) 2 (C) 0 (D) –1

Ans. D

Sol. Required Area $= \int_1^2 \left(\frac{1}{x} - \frac{a}{x^2}\right) dx = \left(\ln x + \frac{a}{x^2}\right) = \ln 2 + \frac{a}{2} - a$



Now, $\ln 2 + \frac{a}{2} - a = \ln 2 - \frac{1}{7}$

$\frac{a}{2} = \frac{1}{7}$ $\Rightarrow$ $7a = 2$ and $7a - 3 = -1$

168. The parabola $y^2 = 4x$ divides the area of the circle $x^2 + y^2 = 5$ in two parts. The area of the smaller part is equal to:

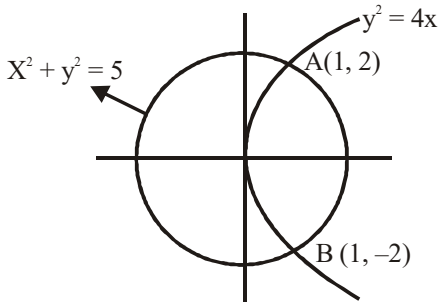
परवलय $y^2 = 4x$, वृत्त $x^2 + y^2 = 5$ के क्षेत्रफल को दो भागों में बाँटता है। छोटे भाग का क्षेत्रफल है :

(A) $\frac{1}{3} + \sqrt{5}\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$ (B) $\frac{1}{3} + 5\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$

(C) $\frac{2}{3} + 5\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$ (D) $\frac{2}{3} + \sqrt{5}\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$

Ans. C

Sol.



$$\text{Area of smaller region} = \int_{-2}^{2}\left(\sqrt{5-y^2} - \frac{y^2}{4}\right)dy$$

$$= 2\int_{0}^{2}\left(\sqrt{5-y^2} - \frac{y^2}{4}\right)dy$$

$$= 2\left[\left(\frac{y}{2}\sqrt{5-y^2} + \frac{5}{2}\sin^{-1}\left(\frac{y}{\sqrt{5}}\right)\right)_0^2 - \frac{1}{12}\left(y^3\right)_0^2\right]$$

$$= 2\left[1 + \frac{5}{2}\sin^{-1}\left(\frac{2}{\sqrt{5}}\right) - \frac{2}{3}\right] = \frac{2}{3} + 5\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$$

169. The area of the region $\left\{(x,y): y^2 \le 4x, x < 4, \frac{xy(x-1)(x-2)}{(x-3)(x-4)} > 0, x \ne 3\right\}$ is :

क्षेत्र $\left\{(x,y): y^2 \le 4x, x < 4, \frac{xy(x-1)(x-2)}{(x-3)(x-4)} > 0, x \ne 3\right\}$ का क्षेत्रफल है :

(A) $\frac{32}{3}$ (B) $\frac{16}{3}$ (C) $\frac{64}{3}$ (D) $\frac{8}{3}$

Ans. A

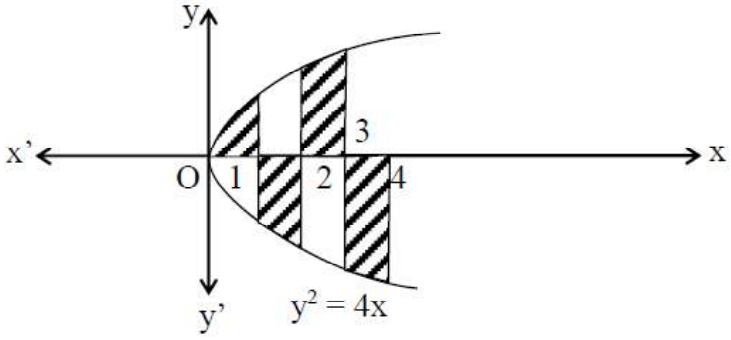
Sol. $\left\{(x,y); y^2 \le 4x, x < 4, \frac{xy(x-1)(x-2)}{(x-3)(x-4)} > 0, x \ne 3\right\}$

$y^2 \le 4x,\ x < 4,\ \frac{xy(x-1)(x-2)}{(x-3)(x-4)} > 0$

Case (i) $y > 0$ $\Rightarrow \frac{x(x-1)(x-2)}{(x-3)(x-4)} < 0$

$x \in (1,2) \cup (3,4)$

$$\text{Area} = \int_0^1 2\sqrt{x}\,dx + \left|\int_1^2 2\sqrt{x}\,dx\right| + \int_2^3 2\sqrt{x}\,dx + \left|\int_3^4 2\sqrt{x}\,dx\right|$$



$$= \frac{4}{3}\left(\left[x^{\frac{3}{2}}\right]_0^1 + \left|\left[x^{\frac{3}{2}}\right]_1^2\right| + \left[x^{\frac{3}{2}}\right]_2^3 + \left|\left(x^{\frac{3}{2}}\right)_3^4\right|\right)$$

$$A = \frac{4}{3}\left(1 + 2\sqrt{2} - 1 + 3\sqrt{3} - 2\sqrt{2} + 4\sqrt{4} - 3\sqrt{3}\right) = \frac{4}{3} \times 8 = \frac{32}{3}$$

170. The area enclosed by the curves $xy + 4y = 16$ and $x + y = 6$ is equal to :

वक्रों $xy + 4y = 16$ तथा $x + y = 6$ से घिरे क्षेत्र का क्षेत्रफल है :

(A) $28 - 30 \log_e 2$ (B) $32 - 30 \log_e 2$ (C) $30 - 28\log_e 2$ (D) $30 - 32 \log_e 2$

Ans. D

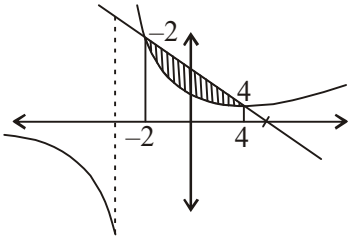
Sol. $y(x + 4) = 16$ & $y = 6 - x$

solve both curves,

$(6 - x)(x + 4) = 16$

$6x - x^2 + 24 - 4x = 16$

$x^2 - 2x - 8 = 0 \Rightarrow x = -2, 4$



$$\int_{-2}^{4}\left(6-x-\left(\frac{16}{x+4}\right)\right)dx$$

$$\Rightarrow\left[6x-\frac{x^2}{2}-16\ell n|x+4|\right]_{-2}^{4}$$

$$\Rightarrow(24-8-16\ell n8)-(-12-2-16\ell n2)$$

$$\Rightarrow 30-32\ell n2$$

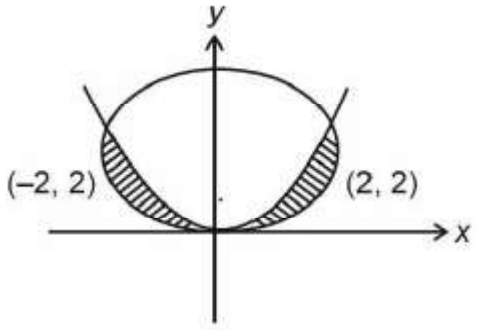
171. Area of the region $\{(x,y):x^2+(y-2)^2\le 4, x^2\ge 2y\}$ is :

क्षेत्र $\{(x,y):x^2+(y-2)^2\le 4, x^2\ge 2y\}$ का क्षेत्रफल :

(A) $2\pi-\frac{16}{3}$ (B) $\pi+\frac{8}{3}$ (C) $\pi-\frac{8}{3}$ (D) $2\pi+\frac{16}{3}$

Ans. A

Sol. $x^2+(y-2)^2\le 4, x^2\ge 2y$



Area of required region $=2\left[\frac{1}{4}\pi(4)-\int_0^2\sqrt{2}\cdot\sqrt{y}dy\right]$

$$\Rightarrow 2\left[\pi-\frac{\sqrt{2}\cdot y^{3/2}}{3/2}\Big|_0^2\right]$$

$$\Rightarrow 2\left[\pi - \frac{2\sqrt{2}}{3}\cdot 2\sqrt{2}\right] = 2\left[\pi - \frac{8}{3}\right] = 2\pi - \frac{16}{3}$$

172. The area of the region enclosed by the curve $y = x^3$ and its tangent at the point $(-1, -1)$ is :

वक्र $y = x^3$ तथा इसके बिंदु $(-1, -1)$ पर स्पर्श रेखा से घिरे क्षेत्र का क्षेत्रफल है :

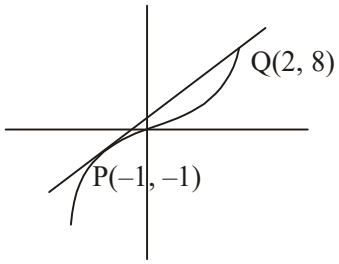
(A) $\frac{23}{4}$ (B) $\frac{19}{4}$ (C) $\frac{27}{4}$ (D) $\frac{31}{4}$

Ans. C

Sol. Given $y = x^3$ _____(1)

$$\Rightarrow \frac{dy}{dx} = 3x^2$$

$$\left(\frac{dy}{dx}\right)_{(-1,-1)} = 3$$



Equation of tangent at $(-1, -1)$

$(y + 1) = 3(x + 1)$

$y = 3x + 2$ _____(2)

Solving (1) and (2)

$x^3 = 3x + 2$

$x^3 - 3x - 2 = 0$ — -1, -1, α

$-2 + \alpha = 0$

$\alpha = 2$

$Q \equiv (2, 8)$

$$\text{Required area} = \int_{-1}^{2}\left(3x + 2 - x^3\right)dx$$

$$= \frac{3}{2}(4-1) + 2(2+1) - \frac{1}{4}(16-1)$$

$$=\frac{9}{2}+6-\frac{15}{4}=\frac{18+24-15}{4}=\frac{27}{4}$$

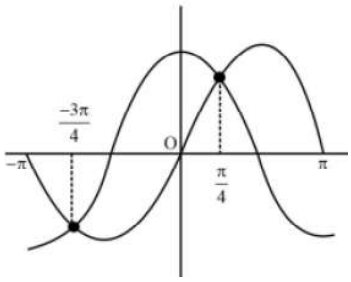
173. The area of the region enclosed by the curve $f(x)=\max\{\sin x,\cos x\}, -\pi\le x\le\pi$ and the x-axis is :

वक्र $f(x)=\max\{\sin x,\cos x\}, -\pi\le x\le\pi$ तथा x-अक्ष से घिरे क्षेत्र का क्षेत्रफल है :

(A) 4 (B) $2(\sqrt{2}+1)$ (C) $2\sqrt{2}(\sqrt{2}+1)$ (D) $4(\sqrt{2})$

Ans. A

Sol.



Area =

$$\left|\int_{-\pi}^{\frac{-3\pi}{4}}\sin x\,dx\right|+\left|\int_{\frac{-3\pi}{4}}^{\frac{-\pi}{2}}\cos x\,dx\right|+\int_{\frac{-\pi}{2}}^{\frac{\pi}{4}}\cos x\,dx+\int_{\frac{\pi}{4}}^{\pi}\sin x\,dx=4$$

174. One of the points of intersection of the curves $y=1+3x-2x^2$ and $y=\frac{1}{x}$ is $\left(\frac{1}{2},2\right)$. Let the area of the region enclosed by the curves be $\frac{1}{24}\left(\ell\sqrt{5}+m\right)-n\log_e\left(1+\sqrt{5}\right)$, where $\ell, m, n\in N$. Then $\ell+m+n$ is equal to :

वक्रों $y=1+3x-2x^2$ तथा $y=\frac{1}{x}$ का एक प्रतिच्छेदन $\left(\frac{1}{2},2\right)$ है। माना इन वक्रों से घिरे का क्षेत्रफल $\frac{1}{24}\left(\ell\sqrt{5}+m\right)-n\log_e\left(1+\sqrt{5}\right)$ है, जहाँ $\ell, m, n\in N$ है, तो $\ell+m+n$ का मान होगा :

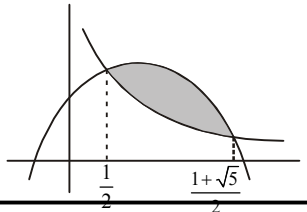
(A) 32 (B) 29 (C) 31 (D) 30

Ans. D

Sol. $2x^3-3x^2-x+1=0$

$\Rightarrow (2x-1)(x^2-x-1)=0$

$\Rightarrow x=\frac{1}{2}, x=\frac{1+\sqrt{5}}{2}$



$$\text{Area} = \int_{\frac{1}{2}}^{\frac{1+\sqrt{5}}{2}} \left(1+3x-2x^2-\frac{1}{x}\right)dx$$

$$= \left[x + + \frac{3x^2}{2} - \frac{2x^3}{3} - \ell n x\right]$$

$$= \frac{\sqrt{5}+1}{2} + \frac{3}{8}\left(\sqrt{5}+1\right)^2 - \frac{1}{12}\left(\sqrt{5}+1\right)^3 - \ell n\left(\frac{\sqrt{5}+1}{2}\right) - \left(\frac{1}{2}+\frac{3}{8}-\frac{1}{12}-\ell n\frac{1}{2}\right)$$

$$= \frac{1}{24}\left[12\left(\sqrt{5}+1\right)+9\left(\sqrt{5}+1\right)^2-2\left(\sqrt{5}+1\right)^3-12-9+2\right] - \ell n\left(\frac{\sqrt{5}+1}{2}\times 2\right)$$

$$= \frac{1}{24}\left[12\left(\sqrt{5}+1\right)+9\left(6+2\sqrt{5}\right)-2\left(5\sqrt{5}+1+3\sqrt{5}\left(\sqrt{5}+1\right)-19\right)\right] - \ell n\left(\sqrt{5}+1\right)$$

$$= \frac{1}{24}\left[14\sqrt{5}+15\right] - \ell n\left(\sqrt{5}+1\right)$$

$\therefore l = 14, m = 15, n = 1$

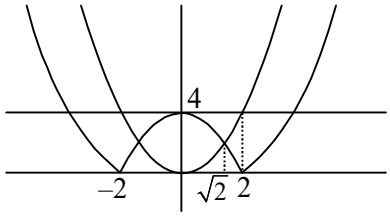
$l + m + n = 30$

175. If the area of the region $\left\{(x,y): |4-x^2| \le y \le x^2, y \le 4, x \ge 0\right\}$ is $\left(\frac{80\sqrt{2}}{\alpha} - \beta\right)$, $\alpha, \beta \in \mathbf{N}$, then $\alpha + \beta$ is equal to_____.

यदि क्षेत्र $\left\{(x,y): |4-x^2| \le y \le x^2, y \le 4, x \ge 0\right\}$ का क्षेत्रफल $\left(\frac{80\sqrt{2}}{\alpha} - \beta\right)$, $\alpha, \beta \in \mathbf{N}$ है, तो $\alpha + \beta$ बराबर है ____.

Ans. 22

Sol.



$$A = \int_0^4 \sqrt{4+y}\,dy - \int_0^2 \sqrt{4-y}\,dy - \int_2^4 \sqrt{y}\,dy$$

$$= \left(\frac{(4+y)^{\frac{3}{2}}}{\frac{3}{2}}\right)_0^4 + \left(\frac{(4-y)^{\frac{3}{2}}}{\frac{3}{2}}\right)_0^2 - \left(\frac{y^{\frac{3}{2}}}{\frac{3}{2}}\right)_0^4$$

$$\frac{80\sqrt{2}}{3} - 16 = \frac{40\sqrt{2}}{3} - 16$$

$\alpha = 6, \beta = 16$

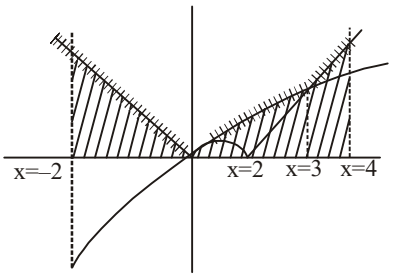
$\alpha + \beta = 22$

176. The area of the region bounded by the curve $y = \max\{|x|, x|x-2|\}$, the x-axis and the lines $x = -2$ and $x = 4$ is equal to _____.

वक्र $y = \max\{|x|, x|x-2|\}$, x-अक्ष और रेखाओं $x = -2$, $x = 4$ से परिबद्ध क्षेत्र का क्षेत्रफल है _____.

Ans. 12

Sol.



$$\text{Req. Area} = \frac{1}{2}\times 2\times 2 + \frac{1}{2}\times 3\times 3 + \frac{1}{2}\times 1\times 11$$

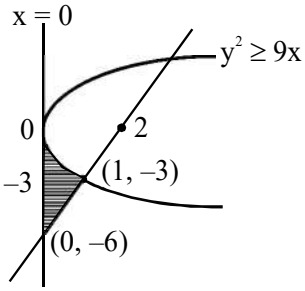
$= 12$

177. Let the area of the bounded region $\{(x, y) : 0 \le 9x \le y^2, y \ge 3x - 6\}$ be A. Then 6A is equal to _____.

माना परिबद्ध क्षेत्र $\{(x, y) : 0 \le 9x \le y^2, y \ge 3x - 6\}$ का क्षेत्रफल A है। तो 6A बराबर है _____.

Ans. 15

Sol.



$$A = \int_{-3}^{0}\left(\frac{y^2}{9} - 0\right)dy + \frac{1}{2}.1.3$$

$$A = \left(\frac{y^3}{27}\right)_{-3}^{0} + \frac{3}{2} = \frac{5}{2}$$

$6A = 15$

178. Let the function,

$$f(x) = \begin{cases} -3ax^2 - 2, & x < 1 \\ a^2 + bx, & x \geqslant 1 \end{cases}$$

be differentiable for all $x \in R$, where $a > 1$, $b \in R$. If the area of the region enclosed by $y = f(x)$ and the line $y = -20$ is $\alpha + \beta\sqrt{3}, \alpha, \beta \in Z$, then the value of $\alpha + \beta$ is _____.

माना कि फलन,

$$f(x) = \begin{cases} -3ax^2 - 2, & x < 1 \\ a^2 + bx, & x \geqslant 1 \end{cases}$$

सभी $x \in R$ के लिए अवकलनीय है, जहाँ $a > 1$, $b \in R$ है। यदि $y = f(x)$ और रेखा $y = -20$ द्वारा परिबद्ध क्षेत्र का क्षेत्रफल $\alpha + \beta\sqrt{3}, \alpha, \beta \in Z$ है, तो $\alpha + \beta$ का मान है _____.

Ans. 34

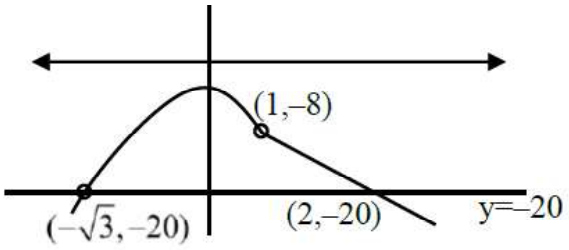
Sol. f(x) is continuous and differentiable

at $x = 1$; LHL = RHL, LHD = RHD

$-3a - 2 = a^2 + b$, $-6a = b$

$a = 2$, $b = -12$

$$f(x) = \begin{cases} -6x^2 - 2; & x < 1 \\ 4 - 12x; & x \geq 1 \end{cases}$$



$$\text{Area} = \int_{-\sqrt{3}}^{1}\left(-6x^2-2+20\right)dx+\int_{1}^{2}\left(4-12x+20\right)dx$$

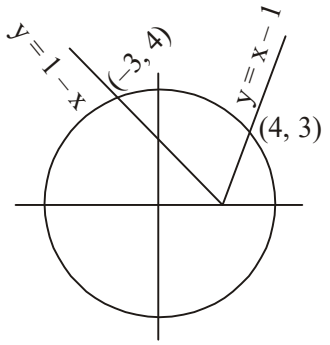
$$16+12\sqrt{3}+6=22+12\sqrt{3}$$

179. If the area of the larger portion bounded between the curves $x^2+y^2=25$ and $y=|x-1|$ is $\frac{1}{4}(b\pi+c)$, b, c $\in$ N, then b + c is equal to _____.

यदि वक्रों $x^2 + y^2 = 25$ और $y = |x - 1|$ के बीच परिबद्ध वृहत्तर भाग का क्षेत्रफल $\frac{1}{4}(b\pi+c)$, b, c $\in$ N है, तो b + c बराबर है _____.

Ans. 77

Sol.



$$A = 25\pi-\int_{-3}^{4}\left(\sqrt{25-x^2}-|x-1|\right)dx$$

$$=25\pi-\left[\frac{1}{2}x\sqrt{25-x^2}+\frac{25}{2}\sin^{-1}\frac{x}{5}\right]_{-3}^{4}+8+\frac{9}{2}$$

$$=\frac{1}{4}(75\pi+2)$$

$\alpha = 75 \quad \beta = 2$

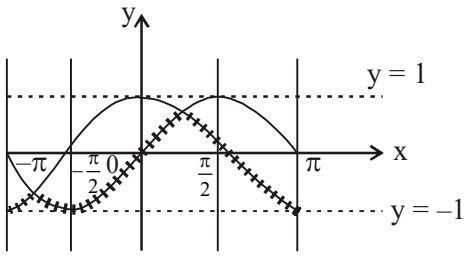
$\alpha + \beta = 77$

180. Let the area of the region enclosed by the curve y = min {sin x, cos x} and the x-axis between $x = -\pi$ to $x = \pi$ be A. Then A^2 is equal to _____.

वक्र y = min {sin x, cos x} तथा x-अक्ष के बीच परिबद्ध क्षेत्र का क्षेत्रफल A है। जहाँ x-अक्ष $x = -\pi$ तथा $x = \pi$ के बीच है, तो A^2 का मान होगा _____

Ans. 16

Sol. $y = f(x) = \min \{\sin x, \cos x\}$



$$A = -\int_{-\pi}^{\frac{-3\pi}{4}} \cos x dx - \int_{\frac{-3\pi}{4}}^{0} \sin x dx + \int_{0}^{\frac{\pi}{4}} \sin x dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos x dx$$

$$-\int_{\frac{\pi}{2}}^{\pi} \cos x dx$$

$A = 4$

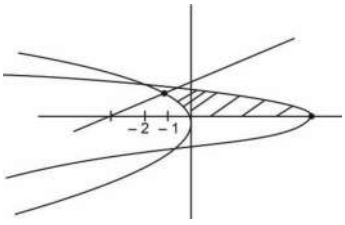
$A^2 = 16$

181. Let the area of the region $\{(x, y) : x - 2y + 4 \geq 0, x + 2y^2 \geq 0, x + 4y^2 \leq 8, y \geq 0\}$ be $\frac{m}{n}$, where m and n are coprime numbers. Then m + n is equal to _____.

माना क्षेत्र $\{(x, y) : x - 2y + 4 \geq 0, x + 2y^2 \geq 0, x + 4y^2 \leq 8, y \geq 0\}$ का क्षेत्रफल $\frac{m}{n}$ है, जहाँ m तथा n असहभाज्य संख्याएँ हैं। तो m + n बराबर है _____।

Ans. 119

Sol.



Area between these curve will be

$$\int_{-2}^{-1}\left(-\sqrt{\frac{-x}{2}} + \frac{x+4}{2}\right)dx + \int_{-1}^{0}\left(\sqrt{\frac{8-x}{4}} - \sqrt{\frac{-x}{2}}\right)dx + \int_{0}^{8}\sqrt{\frac{8-x}{4}}dx = \frac{107}{12} = \frac{m}{n}$$

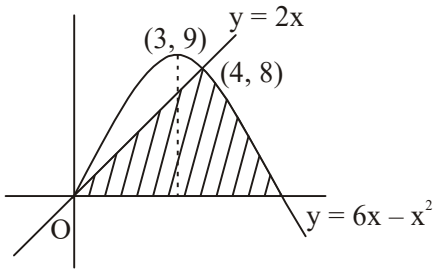
$\Rightarrow m + n = 119$

182. If the area of the region $\left\{(x, y) : 0 \leq y \leq \min\left\{2x, 6x - x^2\right\}\right\}$ is A, then 12 A is equal to _____.

यदि क्षेत्र $\left\{(x,y): 0 \le y \le \min\left\{2x, 6x - x^2\right\}\right\}$ का क्षेत्रफल A है, तो 12 A बराबर है _____

Ans. 304

Sol.



Solving $y = 2x$ and $y = 6x - x^2$

$2x = 6x - x^2$

$x^2 - 4x = 0$

$y = x = 0, x = 4, y = 8$

$A = \frac{1}{2}4.8 + \int_4^6 \left(6x - x^2\right)dx$

$= 16 + \left(3x^2 - \frac{x^3}{3}\right)_4^6$

$A = 16 + 3(36 - 16) - \frac{1}{3}(216 - 64)$

$A = 16 + 60 - \frac{152}{3}$

$A = 16 + \frac{28}{3}$

$12A = 304$

183. The area (in sq. units) of the part of the circle $x^2 + y^2 = 169$ which is below the line $5x - y = 13$ is $\frac{\pi\alpha}{2\beta} - \frac{65}{2} + \frac{\alpha}{\beta}\sin^{-1}\left(\frac{12}{13}\right)$, where α, β are coprime number. Then $\alpha + \beta$ is equal to _____.

वृत्त $x^2 + y^2 = 169$ के रेखा $5x - y = 13$ से नीचे के भाग का क्षेत्रफल (वर्ग इकाई में) $\frac{\pi\alpha}{2\beta} - \frac{65}{2} + \frac{\alpha}{\beta}\sin^{-1}\left(\frac{12}{13}\right)$ है, जहां α, β असहभाज्य संख्याऐं है। तो $\alpha + \beta$ बराबर है_____.

Ans. 171

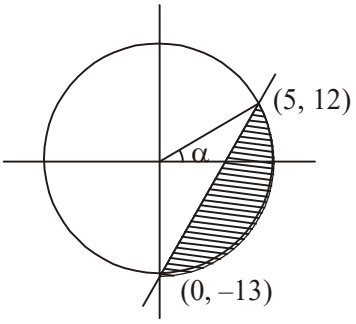
Sol. Solving line and circle $x = 0, 5$

$$\tan\alpha = \frac{12}{5}$$

$$\sin\alpha = \frac{12}{13}$$

$$\sin\left(\frac{\pi}{2}+\alpha\right) = \cos\alpha = \frac{5}{13}$$

Now area of segment $= \frac{1}{2}\left(r^2\right)(\theta - \sin\theta)$



$$= \frac{1}{2}\times 169 \times \left[\frac{\pi}{2} + \sin^{-1}\alpha - \sin\left(\frac{\pi}{2}+\alpha\right)\right]$$

$$= \frac{169}{2}\left[\frac{\pi}{2} + \sin^{-1}\frac{12}{13} - \frac{5}{13}\right]$$

$$= \frac{169\pi}{4} + \frac{169}{2}\sin^{-1}\frac{12}{13} - \frac{65}{2}$$

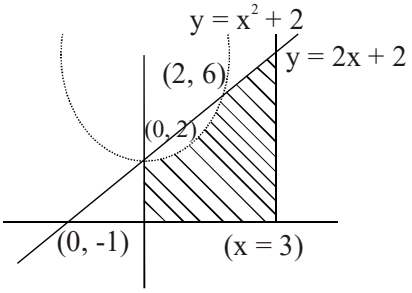
$\alpha = 169$, $\beta = 2$

184. Let the area of the region $\{(x, y) : 0 \le x \le 3, 0 \le y \le \min\{x^2 + 2, 2x + 2\}\}$ be A. Then 12A is equal to :

माना क्षेत्र $\{(x, y) : 0 \le x \le 3, 0 \le y \le \min\{x^2 + 2, 2x + 2\}\}$ का क्षेत्रफल A है। तो 12A बराबर है।

Ans. 164

Sol.



$$A = \int_0^2 \left(x^2 + 2\right)dx + \int_2^3 (2x + 2)dx$$

$$A = \left(\frac{x^3}{3} + 2x\right)_0^2 + \left(x^2 + 2x\right)_2^3$$

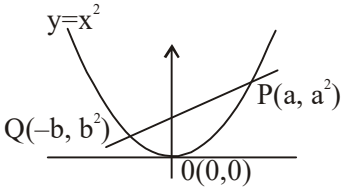
$$A = \frac{41}{3} \Rightarrow 12A = 164$$

185. Three points O(0, 0), P(a, a^2), Q(–b, b^2), a > 0, b > 0, are on the parabola $y = x^2$. Let S_1 be the area of the region bounded by the line PQ and the parabola, and S_2 be the area of the triangle OPQ. If the minimum value of $\frac{S_1}{S_2}$ is $\frac{m}{n}$, gcd(m, n) = 1, then m + n is equal to _____.

परवलय $y = x^2$ पर तीन बिंदु O(0, 0), P(a, a^2), Q(–b, b^2), a > 0, b > 0 हैं। माना रेखा PQ तथा परवलय से घिरे क्षेत्र का क्षेत्रफल S_1 है तथा त्रिभुज OPQ का क्षेत्रफल S_2 है। यदि $\frac{S_1}{S_2}$ का न्यूनतम मान $\frac{m}{n}$, gcd(m, n) = 1 हैं, तो m + n बराबर है _____

Ans. 7

Sol.



Equation of PQ

$$y + b = \frac{a^2 - b^2}{a + b}\left(x - b^2\right)$$

$$\Rightarrow y = (a - b)x + ab$$

$$S_1 = \int_{-b}^{a}\left((a - b)x + ab - x^2\right)dx$$

$$= \left(\frac{(a - b)}{2}x^2 + abx - \frac{x^3}{3}\right)_{-b}^{a}$$

$$= \frac{1}{6}(a + b)^3$$

$$S_2 = \frac{1}{2}\begin{vmatrix} -b & b^2 & 1 \\ 0 & 0 & 1 \\ a & a^2 & 1 \end{vmatrix} = \frac{1}{2}ab(a + b)$$

$$\frac{S_1}{S_2}=\frac{\frac{1}{6}(a+b)^3}{\frac{1}{2}\cdot ab(a+b)}$$

$$=\frac{1}{3}\frac{(a+b)^2}{ab}=\frac{1}{3}\left(\frac{a}{b}+\frac{b}{a}+2\right)$$

$$\because \frac{b}{a}+\frac{a}{b}\geq 2$$

$$\Rightarrow \left(\frac{S_1}{S_2}\right)_{min}=\frac{4}{3}=\frac{m}{n}$$

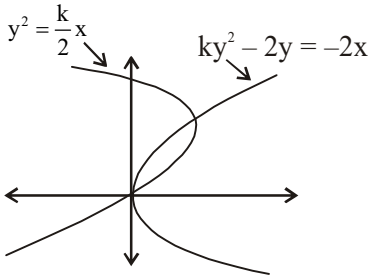
$$\Rightarrow m+n=7$$

186. The sum of squares of all possible values of k, for which area of the region bounded by the parabolas $2y^2 = kx$ and $ky^2 = 2(y - x)$ is maximum, is equal to _____.

k के सभी संभव मानों, जिनके लिए परवलयों $2y^2 = kx$ तथा $ky^2 = 2(y - x)$ से घिरे क्षेत्र का क्षेत्रफल अधिकतम है, के वर्गों का योग है _____

Ans. 8

Sol.



$$\text{Area} = A = \int_0^{\frac{2k}{k^2+4}}\left[\left(\frac{ky^2-2y}{-2}\right)-\left(\frac{2y^2}{k}\right)\right]dy$$

$$\text{Area} = \left[\left(\frac{-k}{2}-\frac{2}{k}\right)\frac{y^3}{3}+\frac{y^2}{2}\right]_0^{\frac{2k}{k^2+4}}$$

$$\text{Area} = \frac{-\left(k^2+4\right)}{2k}\times\left(\frac{2k}{k^2+4}\right)^3\times\frac{1}{3}+\left(\frac{2k}{k^2+4}\right)^2\times\frac{1}{2}$$

$$= -\left(\frac{2k}{k^2+4}\right)^2 \times \frac{1}{3} + \left(\frac{2k}{k^2+4}\right)^2 \times \frac{1}{2}$$

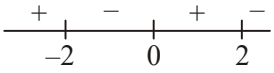
$$= \frac{1}{6} \times \frac{4k^2}{\left(k^2+4\right)^2}$$

$$A(k) = \frac{2}{3}\frac{k^2}{\left(k^2+4\right)^2}$$

$$A'(K) = \frac{2}{3}\frac{2K\left(K^2+4\right)^2 - 2K^2\left(K^2+4\right)(2K)\Big]}{\left(K^2+4\right)^4} = 0$$

$$\Rightarrow \frac{2}{3}(2K)\left(K^2+4\right)\left[K^2+4-2K^2\right] = 0$$

K = 0 or K = 2 or K = –2



$\therefore$ K = 0 is local minima

K = –2 and 2 are local maxima

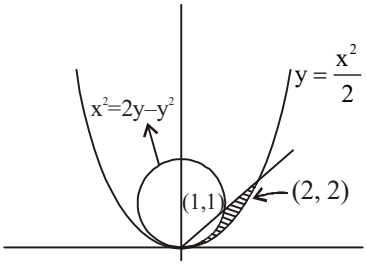
$\therefore (-2)^2 + 2^2 = 8$

187. If the area of the region $S = \left\{(x, y) : 2y - y^2 \le x^2 \le 2y, x \ge y\right\}$ is equal to $\frac{n+2}{n+1} - \frac{\pi}{n-1}$, then the natural number n is equal to _____.

यदि क्षेत्र $S = \left\{(x, y) : 2y - y^2 \le x^2 \le 2y, x \ge y\right\}$ का क्षेत्रफल $\frac{n+2}{n+1} - \frac{\pi}{n-1}$ के बराबर है, तब प्राकृतिक संख्या n बराबर है _____

Ans. 5

Sol. $2y - y^2 \le x^2 \le 2y, x \ge y$



$$\text{Area} = \int_1^2 (\sqrt{2y} - y)dy + \int_0^1 \left(\sqrt{2y} - \sqrt{2y - y^2}\right)dy$$

$$= \frac{7 - 2^{\frac{5}{2}}}{6} + \frac{2^{\frac{7}{2}} - 3\pi}{12}$$

$$= \frac{14 - 2^{\frac{7}{2}} + 2^{\frac{7}{2}} - 3\pi}{12} = \frac{7}{6} - \frac{\pi}{4}$$

$n = 5$

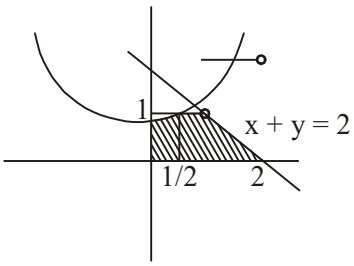
188. Let the area enclosed by the lines $x + y = 2$, $y = 0$, $x = 0$ and the curve $f(x) = \min\left\{x^2 + \frac{3}{4}, 1 + [x]\right\}$ where $[x]$ denotes the greatest integer $\le x$, be A. Then the value of 12A is _____.

माना $[x]$ महत्तम पूर्णांक $\le x$ है। माना रेखाओं $x + y = 2$, $y = 0$, $x = 0$ तथा वक्र $f(x) = \min\left\{x^2 + \frac{3}{4}, 1 + [x]\right\}$ से घिरे क्षेत्र का क्षेत्रफल A है। तो 12 A का मान है :

Ans. 17

Sol. $$A = \int_0^{\frac{1}{2}} \left(x^2 + \frac{3}{4}\right)dx + \frac{1}{2}\left(\frac{3}{2} + \frac{1}{2}\right) \times 2$$

$$\frac{1}{3}.\frac{1}{8} + \frac{3}{4}.\frac{1}{2} + 1$$



$$12A = \frac{1}{2} + \frac{9}{2} + 12 = 17$$

189. Let $y = p(x)$ be the parabola passing through the points $(-1, 0)$, $(0, 1)$ and $(1, 0)$. If the area of the region $\left\{(x, y) : (x + 1)^2 + (y - 1)^2 \le 1, y \le p(x)\right\}$ is A, then $12(\pi - 4A)$ is equal to _____.

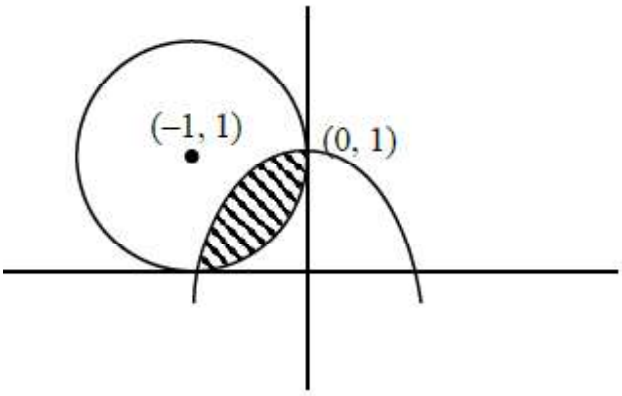
माना बिंदुओं $(-1, 0)$, $(0, 1)$ तथा $(1, 0)$ से होकर जाने वाला परवलय $y = p(x)$ है। यदि क्षेत्र

$\{(x,y):(x+1)^2+(y-1)^2\le 1, y\le p(x)\}$ का क्षेत्रफल A है, तो $12(\pi-4A)$ बराबर है _____

Ans. 16

Sol. There can be infinitely many parabolas through given points.

Let parabola $x^2=-4a(y-1)$



Passes through (1, 0)

$\therefore b=-4a(-1)\Rightarrow a=\frac{1}{4}$

$\therefore x^2=-(y-1)$

Now area covered by parabola $=\int_{-1}^{0}\left(1-x^2\right)dx$

$=\left(x-\frac{x^3}{3}\right)_1^b=(0-0)-\left\{-1+\frac{1}{3}\right\}$

$=\frac{2}{3}$

Required Area = Area of sector – {Area of square – Area covered by Parabola}

$=\frac{\pi}{4}-\left\{1-\frac{2}{3}\right\}$

$=\frac{\pi}{4}-\frac{1}{3}$

$\therefore 12(\pi-4A)=12\left[\pi-4\left(\frac{\pi}{4}-\frac{1}{3}\right)\right]$

$=12\left[\pi-\pi+\frac{4}{3}\right]$

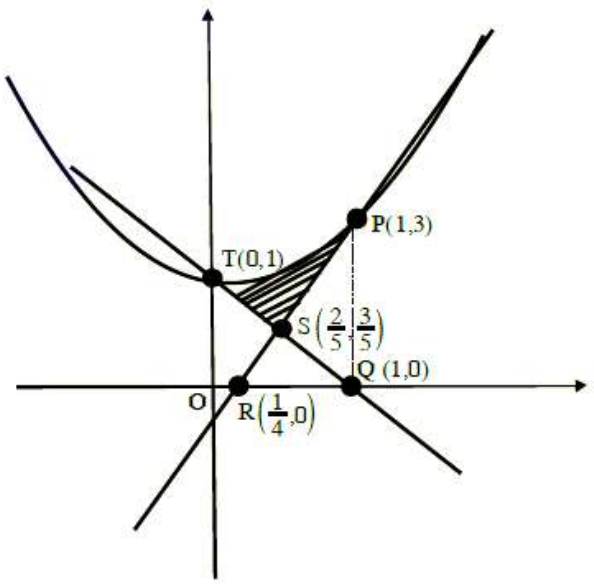
$=16$

190. If A is the area in the first quadrant enclosed by the curve C : $2x^2 - y + 1 = 0$, the tangent to C at the point (1, 3) and the line $x + y = 1$, then the value of 60A is _____.

यदि वक्र C : $2x^2 - y + 1 = 0$, C के बिंदु (1, 3) पर स्पर्श रेखा तथा रेखा $x + y = 1$ से घिरे, प्रथम चतुर्थांश में, क्षेत्र का क्षेत्रफल A है, तो 60A बराबर है _____

Ans. 16

Sol.



$y = 2x^2 + 1$

Tangent at (1, 3)

$y = 4x - 1$

$$A = \int_0^1 \left(2x^2 + 1\right)dx - \text{area of}(\Delta QOT) - \text{are of}(\Delta PQR) + \text{area of}(\Delta QRS)$$

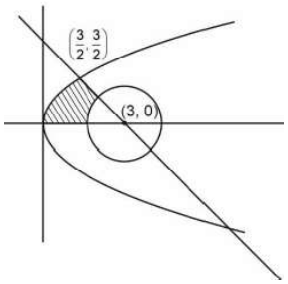
$$A = \left(\frac{2}{3} + 1\right) - \frac{1}{2} - \frac{9}{8} + \frac{9}{40} = \frac{16}{60}$$

191. If the area bounded by the curve $2y^2 = 3x$, lines $x + y = 3$, $y = 0$ and outside the circle $(x - 3)^2 + y^2 = 2$ is A, then $4(\pi + 4A)$ is equal to _____.

यदि वक्र $2y^2 = 3x$, रेखाओं $x + y = 3$, $y = 0$ तथा वृत्त $(x - 3)^2 + y^2 = 2$ के बाहर गिरा हुआ क्षेत्रफल A है, तो $4(\pi + 4A)$ मान होगा _____

Ans. 42

Sol.



$$\int_0^{3/2}\left((3-y)-\frac{2y^2}{3}\right)dy-\pi\left(\sqrt{2}\right)^2\cdot\frac{1}{8}$$

$$=\frac{36-9-6}{8}-\frac{\pi}{4}=\frac{21}{8}-\frac{\pi}{4}$$

$4(\pi + 4A) = 42$